SVM

INVERTER

Current Vector Control (RPI based)

$\vec{u}_{\alpha\beta}^{ctrl}(t)$

u_a^{ctrl}

u_b^{ctrl}

u_c^{ctrl}

$\vec{i}_s^{abc}\big|^{ref}(t)$

$\vec{i}_s^{abc}(t)$

$\vec{u}_g^{abc}(t)$

DQ-PLL

$\hat{\vartheta}_g(t)$

u_{dc}^{ref} + −

DClink Voltage Control (PI based)

$i_p^{\xi}\big|_s^{ref}(t)$

$u_{dc}(t)$

$\vec{i}_s^{abc}\big|^{ref}(t)$

$\hat{\vartheta}_g(t)$

Sequence Calculation and Inner Current Reference Generator

$\vec{i}_{\xi\eta}^{p}(t)$

$\vec{i}_{\xi\eta}^{n}(t)$

$\vec{u}_{\xi\eta}^{p}(t)$

$\vec{u}_{\xi\eta}^{n}(t)$

dq_p $\alpha\beta$

dq_n $\alpha\beta$

$\alpha\beta$ abc

$\vec{i}_g^{abc}(t)$

$u_{dc}(t)$

$\vec{u}_g^{abc}(t)$